

# Library Management System

## 1. Introduction

A Library Management System is designed to manage and organize the daily operations of a library. It helps in maintaining records of books, members, issue and return transactions efficiently.

## 2. Objective

- To maintain book records
- To manage library members
- To track issued and returned books
- To reduce manual work

## 3. Technologies Used

- SQL (Oracle)
- ER Diagram Design Tools (Draw.io / dbdiagram.io)

## 4. System Modules

1. Book Management
2. Member Management
3. Issue/Return Management

## 5 .Entities:

- Books
- Members
- Issue

### Relationships:

- A member can issue multiple books
- A book can be issued by multiple members over time

## 6. Database Tables

### **Books Table**

- Book\_ID (Primary Key)
- Title
- Author
- Publisher
- Year

### **Members Table**

- Member\_ID (Primary Key)
- Name
- Address
- Phone

### **Issue Table**

- Issue\_ID (Primary Key)
- Book\_ID (Foreign Key)
- Member\_ID (Foreign Key)
- Issue\_Date
- Return\_Date

## **7. SQL Code**

### **Create Books Table**

```
CREATE TABLE Books (  
Book_ID INT PRIMARY KEY,  
Title VARCHAR(50),  
Author VARCHAR(50),  
Publisher VARCHAR(50),  
Year INT  
);
```

### **Create Members Table**

```
CREATE TABLE Members (  
Member_ID INT PRIMARY KEY,  
Name VARCHAR(50),
```

```
Address VARCHAR(100),  
Phone VARCHAR(15)  
);
```

### **Create Issue Table**

```
CREATE TABLE Issue (  
Issue_ID INT PRIMARY KEY,  
Book_ID INT,  
Member_ID INT,  
Issue_Date DATE,  
Return_Date DATE,  
FOREIGN KEY (Book_ID) REFERENCES Books(Book_ID),  
FOREIGN KEY (Member_ID) REFERENCES Members(Member_ID)  
);
```

### **8. Sample Queries**

- View all books  
SELECT \* FROM Books;
- Issue a book  
INSERT INTO Issue VALUES (1, 101, 201, SYSDATE, NULL);
- Return a book  
UPDATE Issue SET Return\_Date = SYSDATE WHERE Issue\_ID = 1;

### **9. Advantages**

- Easy data management
- Reduces paperwork
- Fast access to information

### **10. Conclusion**

The Library Management System simplifies library operations by automating record keeping and improving efficiency. It is useful for schools, colleges, and public libraries.